

Review

Harnessing 'phage training' to bolster the therapeutic potential of bacteriophages

Natasha Torriero-Smith¹, Benjamin A Rogers^{2,3,4},
Michael J McDonald^{1,4} and Jeremy J Barr^{1,4}

Bacteriophages (phages) are viruses that selectively prey on bacteria. Their use in treating antimicrobial-resistant bacterial infections is steadily increasing due to the need for alternative therapies. The application of phage therapy is not without its challenges, including difficulties associated with isolating phages against a target strain, the limited infectivity of a phage, the cost and complexity of producing well-characterised phage stocks, and the emergence of phage resistance. The directed adaptation of phage to a specific bacterial target, also known as 'phage training', leverages the natural evolutionary capacity of phages and can be used to bolster their bacterial killing abilities. Phage training dates back almost as far as phage therapy itself, being used to expand the therapeutic use of phages. Numerous reports showcase the success and benefits of phage training *in vitro* and its potential to operate effectively within the framework of phage therapy. However, the time needed to train a given phage, followed by genotypic and phenotypic characterisation of both pre- and post-trained phages, is a major limitation. Here, we explore oversights of the phage training process and propose some considerations and solutions to help drive the field forward to enable its feasible integration into phage therapy.

Addresses

¹ School of Biological Sciences, Monash University, 25 Rainforest Walk, Clayton 3800, Victoria, Australia² Monash University, School of Clinical Sciences at Monash Health, 246 Clayton Rd, Clayton 3168, Victoria, Australia³ Monash Infectious Diseases, Monash Health, 246 Clayton Rd, Clayton 3168, Victoria, Australia⁴ Centre to Impact AMR, Monash University, Wellington Rd, Clayton 3800, Victoria, AustraliaCorresponding author: Barr, Jeremy J (jeremy.barr@monash.edu)

Current Opinion in Microbiology 2026, 89:102696

This review comes from a themed issue on **Viruses**Edited by **Joe Pogliano**For complete overview of the section, please refer to the article collection, "[Host-Microbe Interactions: Viruses \(2025\)](#)"

Available online 17 December 2025

<https://doi.org/10.1016/j.mib.2025.102696>1369–5274/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Introduction

Antimicrobial resistance

Antimicrobial resistance (AMR) continues to pose a critical global health problem, with an estimated 4.71 million deaths associated with bacterial AMR infections in 2021 [1]. Phage therapy, the clinical use of phages to treat bacterial infections, is gaining significant attention as an alternative or complementary treatment to antibiotics, particularly when standard therapies fail [2,3]. Phages are abundant in natural environments, and are routinely isolated from sources including soil, oceans, and wastewater using target bacteria [4]. Virulent phages that rapidly infect and kill bacteria are preferred for phage therapy. After a phage has been isolated by culture on a clinical strain, characterising the phage's physical, genetic, and biochemical properties is required, but it takes considerable time, labour, and resources [5,6]. Large-scale batch production of candidate phages that are sufficiently pure to administer to patients is also complex and expensive [7–9]. Aside from the practical difficulties in preparing a phage for therapy, additional hurdles exist that stem from the biology of the phages and bacteria themselves. First, phages are highly specific to their bacterial hosts, typically only targeting a small collection of bacterial strains within their target species. Phages are therefore not typically broad-spectrum [10]. Second, the infectivity of available phages against the target bacterial strain may be poor, limiting suitability for therapy. Finally, bacteria readily evolve phage resistance in response to phage predation, potentially rendering phages ineffective in killing the bacteria after a short period of time [11]. These challenges present considerable barriers to the clinical viability of phages.

'Phage training', broadly, is the selective adaptation of phages against target bacteria to improve their therapeutic potential. Trained phages have reported significant bactericidal improvements over their ancestral counterparts, including expanded host range [12–19], improved infectivity [20–22], and the capacity to circumvent or delay bacterial phage resistance [23–26]. While methods for artificial phage modification and engineering exist [27,28], 'natural' training approaches exploit the genetic variation that arises naturally during phage propagation, and benefit from the ability of

evolution to find adaptive innovations that can be difficult for genetic engineers to predict. As ‘trained phages’ are derived from naturally occurring phages that have undergone natural evolutionary pathways, these phages do not face the regulatory issues encountered by genetically modified organisms, and can be more readily approved for phage therapy [29]. In this perspective, we summarise common phage training techniques, their benefits and limitations, and propose important elements to consider when undertaking phage adaptation experiments to bolster candidate phages for therapy.

Can phage training become a standardised part of phage therapy?

Benefits of implementing phage training into the therapeutic pipeline

Unlike antibiotics, phages are biological entities that naturally evolve alongside bacteria [23]. The resulting co-evolutionary ‘arms-race’ between phages and bacteria has shaped their evolution across billions of years [30,31]. It is the inherent capacity to undergo rapid genetic changes to renew or expand their infectivity that makes phage training an ideal process to tailor phages with improved phenotypes. The process itself is simple and inexpensive; currently, it is easier to leverage the natural adaptability of phage genomes than to conduct more elaborate genetic engineering approaches [28,32]. Still, the knowledge gained from ‘natural’ phage training studies will greatly inform phage engineering practices [25,29]. Phages may continue to evolve and maintain infectivity against bacteria after clinical administration. However, the consequential evolution of phage resistance in patients following therapy, and subsequent counter-evolution of phages, remains underexplored [20,29,33]. The use and implementation of phage training is expanding, with more studies reporting expanded phage host range, improved phage infectivity, and overcoming phage resistance while uncovering deeper insights into phage genetics and evolution.

Prevailing phage training methods

Phage adaptation is mainly achieved through *in vitro* directed evolution experiments using select phage–host pairs (Figure 1). Phage training entails the co-incubation of phages with target bacteria, followed by a chosen number of serial transfers. Three broad approaches to phage training have been reported across the literature. The first approach, known as phage evolution, is to evolve only the phages, while suppressing host evolution (Figure 1a). Following incubation, the adapted phage population is recovered from co-culture and transferred to a fresh vessel with evolutionarily naïve bacteria (i.e. not previously subjected to co-culture with phage) to continue the iterative adaptation against the target strain [8,21,29]. This is often done with a single phage–host pair or against multiple bacterial strains to improve phage infectivity and host range.

In a second approach, phage-bacterial co-evolution, a co-evolutionary training method, is used to evolve a phage while also allowing its bacterial host to co-evolve, and is most commonly used to overcome phage resistance (Figure 1b). Phages exploit structures on the bacterial surface to initiate infection, and bacteria in turn can acquire mutations that alter these surface structures to block phage infection. During co-evolutionary experiments, phages may counter-evolve mechanisms to regain infectivity towards the resistant host [23,25,26,31,34]. This typically occurs through alterations to their tail structures to target a different or altered bacterial receptor, which can modify phage host range and delay the emergence of phage-resistance [15,19,35,36].

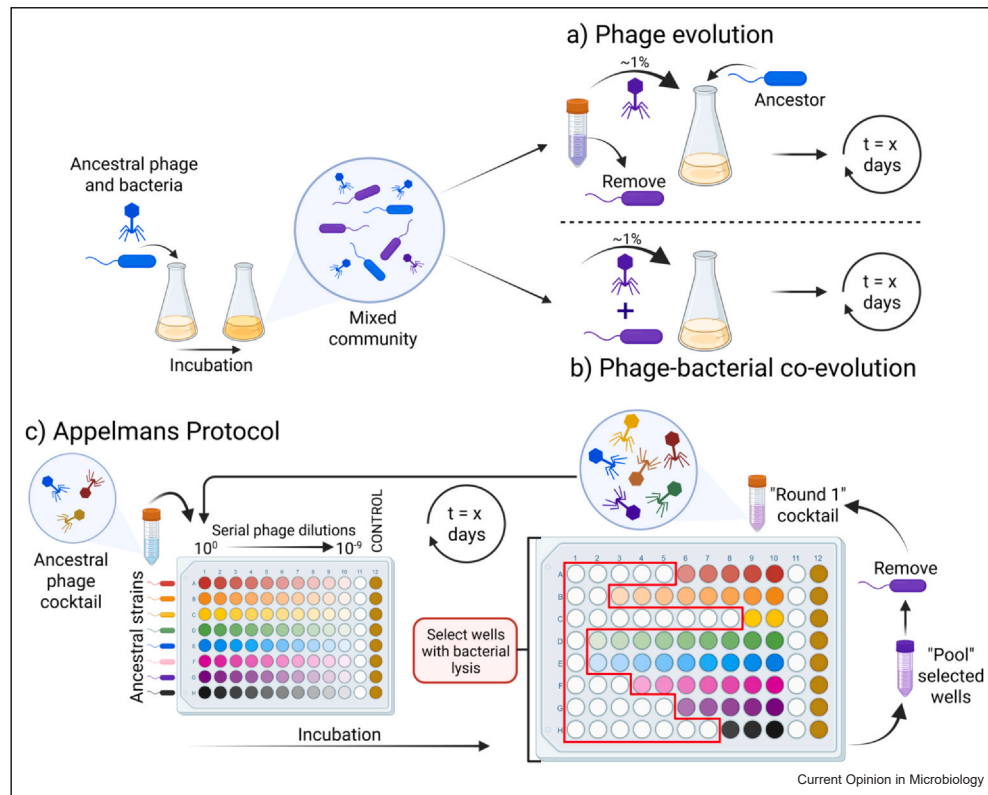
The final approach (Figure 1c) involves broad cross-infection of multiple phages and hosts using the Appelmans protocol [37]. Originally reported in 1921 as a phage titration method, the Appelmans protocol was later adapted for phage training. At its core, multiple phage species are mixed together and then serially diluted, typically in a 96-well plate format, with bacterial strains of interest. Following incubation, each dilution is checked for lysis compared with a control; those wells exhibiting lysis are pooled together regardless of strain. The resulting mixture is cleared of bacteria, and the pooled phages are used to repeat the process until ~80% of the bacterial strains are lysed at an adequately low titre (typically dilution of 10^{-7}). The Appelmans protocol has been adapted in numerous studies published in the last decade to facilitate host range expansion of a single phage or phage cocktail against a broad suite of hosts [12,14,16–18,38–41].

These three primary phage training approaches typically operate for ~30 days [12,14,18,19,23,26,41,42], yet sometimes less [17,22,25] or more [16]. Timeframe is the primary metric used to determine the experimental duration, given the complexities associated with measuring defined phage ‘generations’ beyond a single replication cycle. The evolved phage populations from selected timepoints, usually the experimental endpoint, are then studied further. Phage populations can be screened for the desired traits, followed by metagenomic analyses to track genetic changes in the evolving phage population over time. If the tracking and isolation of individual phage genotypes are desired, phages from these populations are isolated and screened to detect improved phenotypes of interest. This is followed by genomic analyses to uncover genetic variations responsible for the changed phenotypes in the evolved phages relative to the ancestor(s) [12,16,18,24,34,38].

Phage training success stories

Phages may have the capacity to infect and kill a bacterial strain, but be poorly adapted against it. A low efficiency of plating (EOP) means a phage replicates to a

Figure 1



Overview of the common *in vitro* phage training approaches. The phage and the target bacterial strain are co-cultured together. Following incubation, the co-culture contains a mixed community of ancestral (blue) and evolved (purple) phages and bacteria. **(a)** Phage evolution. The bacteria are removed following incubation, and a small portion of the evolving phage population (purple) is transferred to a fresh vessel with nutrient media and the ancestral host (blue), to iteratively passage only the phage. **(b)** Phage-bacterial co-evolution. A small portion of the evolving phage and bacteria (purple) is transferred together to a fresh vessel with nutrient media. **(c)** The Appelmans protocol: 96-well plate format. A phage cocktail is serially diluted by column across the plate. Each row is seeded with different bacterial strains. Following incubation, all wells in each dilution series exhibiting complete bacterial lysis plus the first turbid well are pooled together. The bacteria are removed, and the 'round 1' cocktail is used for the next evolution cycle, following the previous setup.

(c) First adapted by Burrowes et al. [12].

low titre against a target strain relative to a reference strain and may be ineffective for clinical use [43]. Iterative adaptation of a phage against one or more bacterial hosts may improve phage infectivity and replication. A recent study by Subedi et al. [8] improved the lytic capabilities of three phages against a clinical collection of *Enterobacter cloacae* complex (ECC), using phage evolution (Figure 1a). Each phage was passaged with a single host against which they exhibited low to moderate EOP, with one of the three phages trained against a nonpermissive host. The phages were propagated for 24 hours against their respective hosts and were passaged accordingly for 10 consecutive days, showing significant improvements to phage infectivity between five and seven days. The phages showed a significant increase in EOP against their training hosts, including the phage initially unable to productively infect its training host.

A combination of phage training approaches can be applied to drive and compare changes to phage infectivity. This was recently demonstrated by Ngiam et al. [26], who compared evolutionary and co-evolutionary phage training on phage infectivity and overcoming phage-resistance in *Klebsiella pneumoniae* for 30 days. Phages were either continuously passaged against the ancestral strain or co-evolved with both phage and bacteria being transferred. The infectivity, genetics, and phenotypic features of the evolved and co-evolved phages were compared to those of the ancestral phage by studying the mixed phage communities at different timepoints, and two isolated phages, one each from the evolved and co-evolved communities. Both phage training approaches improved phage infectivity, but the use of co-evolution was favoured due to these phages' improved bacterial growth suppression and superior ability to delay the emergence of phage-resistant bacteria. With these

findings, it may be ideal to use phage-bacterial co-evolution (Figure 1b), which entails an easier daily experimental set-up, for improving phage infectivity and delaying the emergence of phage-resistant bacteria.

Host range is one of the foremost limiting factors of phage therapy, and there is a strong incentive to broaden the host range of naturally occurring phages. Expanding phage host range has been achieved primarily through variations of the Appelmans protocol (Figure 1c) [12,16–18,38–40]. The most influential example of the modern Appelmans protocol to date is that described by Burrowes, Molineux, and Fralick [12], who attribute their method to the Eliava Phage Institute in Tbilisi, Georgia [43]. They applied a three-phage cocktail against ten *Pseudomonas aeruginosa* strains to expand the cocktail's host range over 30 days. A phage capable of infecting all 10 strains was isolated, and its genetic origins were traced to at least 48 genetic crossover events between two of the three input phages. They assert that due to the overwhelming prevalence of cross-over events and a lack of single-point mutations, the Appelmans protocol operated primarily via homologous recombination to facilitate the evolution of broader host range phages [12,43]. Later studies that adapted this approach usually (but not always [16]) successfully broadened phage host range, but through varying evolutionary pathways. Beyond recombination, single-point mutations, insertions, and deletions were also deduced to be associated with host range expansion. Notably, the Appelmans protocol has also been adapted to increase phage titre and infectivity [22,38,44] and enhance biofilm degradation [41], showcasing its widespread applicability.

How can we improve phage training?

Phage training obstacles

Phage training experiments indicate substantial promise in improving phage therapeutic potential. However, we highlight two consistent issues to address in future studies.

The first is a lack of standardisation in both procedure and timeframe. Broadly, phage training is a relatively straightforward and low-cost process that can yield improved phages or phage communities accordingly (Figure 1). The methods are flexible in the number and composition of phages and bacteria included, provided there is at least one susceptible host to enable phage propagation. However, the subsequent isolation and characterisation of evolved phages takes substantial time and labour, and it is common for a small collection of phages evolved against one or more strains to quickly become an unruly mix of unknown phage genotypes. Bioinformatic analyses can track these evolutionary changes, yet published reports occasionally lack detail or

depth here. Additionally, there is a lack of translational studies that compare the clinical utility of trained phages with their ancestral counterparts [20,25]. Future studies should first streamline the experimental setup and optimise the timeframe of the phage training pipeline, be aware of the scale and depth required for downstream analysis, and consider comparative models to test the clinical value of phage training.

The second is whether to use evolved communities of phages or instead to isolate singular phage genotypes from these evolved communities. In this context, we define an 'evolved phage community' as a mixture containing phages of different species or variants within that species. These communities will almost certainly consist of genetically different phages, even if the experiment began with a single phage genotype. Comparatively, individual phage genotypes are those that are separately isolated from these mixed communities via standard plaque purification techniques [45]. Many studies focus on evolved communities, while others plaque-purify singular phages, usually from the experimental endpoint [14–16,19,22,26,38,41]. Of those that screen for individual phages, the number of phages isolated can vary substantially, with every phage isolated and characterised adding significant time and complexity to the process. Comparatively, characterising evolved communities of phages is faster, and the frequency of mutations within the community can be tracked over time, yet this approach fails to link genotype with phenotype and omits much of the biological complexity. Another issue to consider when working with evolved phage communities is whether these communities are applicable for translational phage therapy. Given the growing requirement to perform phenotypic and genomic characterisation of each phage before patient administration, it is unclear whether these community approaches are suitable in the developing phage therapy framework.

Rules of phage training

To better optimise and incorporate phage training into the broader phage therapy framework, we propose that future studies consider the following points.

1. *Align adaptive strategy with clinical goals:* Determine what improvements to the phages are required for clinical needs and adapt them accordingly. If the aim is to improve the infectivity of a phage against a patient strain, a short phage evolution protocol (Figure 1a) to repeatedly passage the phage and adapt it to the target strain without co-evolution, followed by isolating and characterising individual phage genotypes, is sufficient. If overcoming phage resistance is the goal, co-evolution (Figure 1b) may 'pre-adapt' the phage to anticipate the evolutionary pathway of its host, improving its ability to kill phage-resistant

bacteria [35]. If broadening phage host range is the intent, then applying the Appelmans protocol for at least 10 days followed by extensive phage isolation and screening against a clinical host panel is likely required (Figure 1c). However, caution should be taken when working with several phage and bacterial species, as this can greatly expand and complicate the downstream workload. The Appelmans protocol is therefore unlikely to be ideal for use in time-critical clinical cases, but could be applied to previously isolated phages to optimise them in preparation for future cases.

2. *Clinical applications require characterised phages:* We assert that clinically applied ‘trained’ phages must be well defined at the genetic and phenotypic levels, and that this can only be done with individual phages. Phage communities may perform better and are easier to validate as they remove the lengthy phage isolation and characterisation process. However, depending on the size and composition of the input phage pool, an evolved phage community may contain hundreds of genotypes, becoming an undefined mixture. Current phage therapy regulations require phage products to be thoroughly characterised, which is an infeasible undertaking within these undefined communities. Communities are also harder to mass-produce as the composition would likely change depending on the propagation host.

Concluding remarks

Phage training is a promising tool that we can leverage to improve phage host range, infectivity, and delay the emergence of phage-resistant bacteria. Before undertaking phage training, we suggest characterising the genomes of the input phage(s) and bacterial strain(s) along with the host range, plaquing efficiency, and the target receptor for each phage. Training a single phage or a mixture of divergent phages against one or more strains is likely to select for mutations in the receptor-binding proteins, known to dictate phage host range [46]. Following this, the desired number of phages to isolate, purify, and characterise should be carefully considered, given the time and complexity required for analysis. Comparative genomics of the evolved phages compared with their ancestral counterparts is essential to link genotype with phenotype. Previous findings suggest that phages are likely to evolve via either recombination and/or point mutations, which can shortlist the bioinformatic approaches required to analyse phage evolution. If prophages are detected in the bacterial hosts, these should be factored into any genetic analysis as they may influence phage evolution and phage training outcomes [16,47]. Once the candidate phage has been genetically characterised, it can then be produced for clinical use.

The outpouring of recent literature examining different phage training approaches indicates that these experiments

are simple to set up and usually effective in improving phages in diverse ways. The genetic mechanisms underlying them are increasingly understood, suggesting that phage evolution can be anticipated and is repeatable [35,38]. However, there are underappreciated costs associated with phage training approaches, including the lengthy timeframe, lack of standardisation between protocols, inconsistency between using evolved phage communities or singular phage genotypes, and occasionally insufficient characterisation of the phages and bacteria. Care must be taken when selecting the phage training approach, the scale of the experiment, and in streamlining phage isolation, characterisation, and analysis. A fast, flexible application of phage training tailored to patient needs may ultimately be the way forward in adding this valuable technique to the phage therapy pipeline.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors have nothing to declare.

Acknowledgements

This work and the efforts by J.J.B. were supported by a National Health and Medical Research Council Investigator Grant Leadership Level 1 (NHMRC; 2026130). B.A.R. is supported by an NHMRC Emerging Leadership Grant (2017661).

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- of special interest
 - of outstanding interest
1. Naghavi M, et al.: **Global burden of bacterial antimicrobial resistance 1990–2021: a systematic analysis with forecasts to 2050.** *Lancet* 2024, **404**:1199–1226.
 2. Sawa T, Moriyama K, Kinoshita M: **Current status of bacteriophage therapy for severe bacterial infections.** *J Intensive Care* 2024, **12**:44.
 3. Würstle S, et al.: **Optimized preparation pipeline for emergency phage therapy against *Pseudomonas aeruginosa* at Yale University.** *Sci Rep* 2024, **14**:2657.
 4. Hyman P: **Phages for phage therapy: isolation, characterization, and host range breadth.** *Pharmaceuticals* 2019, **12**:35.
 5. Gordillo Altamirano FL, Barr JJ: **Unlocking the next generation of phage therapy: the key is in the receptors.** *Curr Opin Biotechnol* 2021, **68**:115–123.
 6. Strathdee SA, et al.: **Phage therapy: from biological mechanisms to future directions.** *Cell* 2023, **186**:17–31.
 7. Khatami A, et al.: **Standardised treatment and monitoring protocol to assess safety and tolerability of bacteriophage therapy for adult and paediatric patients (STAMP study): protocol for an open-label, single-arm trial.** *BMJ Open* 2022, **12**:e065401.
 8. Subedi D, et al.: **Rational design of a hospital-specific phage cocktail to treat *Enterobacter cloacae* complex infections.** *Nat Microbiol* 2025, **10**:2702–2719 p. 2024.06.30.601436.

This publication created a phage cocktail against a large clinical collection of ECC isolates. The initial three-phage product was optimised through iterative phage adaptation to significantly enhance its killing efficiency and host range, forming a new therapeutic-grade cocktail, Entelli-02.

9. Abdelsattar A, et al.: **Bacteriophages: from isolation to application.** *Curr Pharm Biotechnol* 2022, **23**:337–360.
 10. Chung KM, Liao XL, Tang SS: **Bacteriophages and their host range in multidrug-resistant bacterial disease treatment.** *Pharmaceuticals* 2023, **16**:1467, <https://doi.org/10.3390/ph16101467>
 11. Hasan M, Ahn J: **Evolutionary dynamics between phages and bacteria as a possible approach for designing effective phage therapies against antibiotic-resistant bacteria.** *Antibiotics* 2022, **11**:915.
 12. Burrowes BH, Molineux IJ, Fralick JA: **Directed in vitro evolution of therapeutic bacteriophages: the Appelmans Protocol.** *Viruses* 2019, **11**:241.
 13. Subramanian S, et al.: **Host range expansion of Shigella phage Sf6 evolves through point mutations in the tailspike.** *J Virol* 2022, **96**:e00929–22.
 14. Mapes AC, et al.: **Development of expanded host range phage active on biofilms of multi-drug resistant *Pseudomonas aeruginosa*.** *Bacteriophage* 2016, **6**:e1096995.
 15. Boon M, et al.: **Host range expansion of *Pseudomonas* virus LUZ7 is driven by a conserved tail fiber mutation.** *Phage* 2020, **1**:87–90.
 16. Vu TN, et al.: **Appelmans protocol — a directed in vitro evolution enables induction and recombination of prophages with expanded host range.** *Virus Res* 2024, **339**:199272.
- This study was the first reported to apply the Appelmans protocol to broaden the host range of phages against carbapenem-resistant *Acinetobacter baumannii* (CRAB). The resulting phages with broadened host range were identified as recombinant derivatives of prophages induced from the hosts. Their host ranges were also unstable, rendering them unsuitable for therapy.
17. Lossouarn J, et al.: **The virtue of training: extending phage host spectra against vancomycin-resistant *Enterococcus faecium* strains using the Appelmans method.** *Antimicrob Agents Chemother* 2024, **68**:e01439–23.
 18. Kolenda C, et al.: **Potential of training of anti-*Staphylococcus aureus* therapeutic phages against *Staphylococcus epidermidis* multidrug-resistant isolates is restricted by inter- and intra-sequence type specificity.** *mSystems* 2024, **9**:e00850–24.
 19. Ghatbale P, et al.: **Experimental phage evolution results in expanded host ranges against antibiotic resistant *Klebsiella pneumoniae* isolates.** *Nat Commun* 2025, **16**:p. 9903.
 20. Morello E, et al.: **Pulmonary bacteriophage therapy on *Pseudomonas aeruginosa* cystic fibrosis strains: first steps towards treatment and prevention.** *PLoS One* 2011, **6**:e16963.
 21. Betts A, et al.: **Back to the future: evolving bacteriophages to increase their effectiveness against the pathogen *Pseudomonas aeruginosa* PAO1.** *Evol Appl* 2013, **6**:1054–1063.
 22. Kok DN, et al.: **In vitro evolution to increase the titers of difficult bacteriophages: RAMP-UP protocol.** *Phage* 2023, **4**:68–81.
- In this study, the Appelmans protocol was applied to improve replication and titre of a phage against a single host. In just four days, the phage successfully achieved a 100-fold to over 100 000-fold improvement in titre. This approach was re coined RAMP-UP (rapid adaptive mutation of phage-UP).
23. Borin JM, et al.: **Coevolutionary phage training leads to greater bacterial suppression and delays the evolution of phage resistance.** *Proc Natl Acad Sci USA* 2021, **118**:e2104592118.
 24. Borin JM, et al.: **Comparison of bacterial suppression by phage cocktails, dual-receptor generalists, and coevolutionarily trained phages.** *Evol Appl* 2023, **16**:152–162.
 25. Wang M, et al.: **Coevolutionary phage training and Joint application delays the emergence of phage resistance in *Pseudomonas aeruginosa*.** *Virus Evol* 2023, **9**:vead067.
- In this publication, an N4-like phage was co-evolved with a strain of *Pseudomonas aeruginosa*. The evolved phage isolated showed renewed infectivity against the resistant host, and application of the evolved and ancestral phages showed improved killing efficiency and bacterial suppression both *in vitro* and *in vivo*.
26. Ngiam L, Weynberg K, Guo J: **Evolutionary and co-evolutionary phage training approaches enhance bacterial suppression and delay the emergence of phage resistance.** *ISME Commun* 2024, **4**:ycae082.
- In this publication, a phage underwent both evolutionary and co-evolutionary training against *Klebsiella pneumoniae* to compare the effects of both approaches on the phage's genetics and phenotype. Both approaches improved phage infectivity; however, co-evolution demonstrated a superior potential to overcome the phage resistance barrier.
27. Abdelsattar AS, et al.: **How to train your phage: the recent efforts in phage training.** *Biologics* 2021, **1**:70–88, <https://doi.org/10.3390/biologics1020005>
 28. Mahler M, et al.: **Approaches for bacteriophage genome engineering.** *Trends Biotechnol* 2023, **41**:669–685.
 29. Rohde C, et al.: **Expert opinion on three phage therapy related topics: bacterial phage resistance, phage training and prophages in bacterial production strains.** *Viruses* 2018, **10**:178.
 30. Azam AH, Tanji Y: **Bacteriophage-host arm race: an update on the mechanism of phage resistance in bacteria and revenge of the phage with the perspective for phage therapy.** *Appl Microbiol Biotechnol* 2019, **103**:2121–2131.
 31. Burmeister AR, et al.: **Sustained coevolution of phage Lambda and *Escherichia coli* involves inner — as well as outer-membrane defences and counter-defences.** *Microbiology* 2021, **167**:001063.
 32. Bull JJ, Krone SM: **Mathematical comparison of protocols for adapting a bacteriophage to a new host.** *Virus Evol* 2024, **10**:veae100.
 33. Oromí-Bosch A, Antani JD, Turner PE: **Developing phage therapy that overcomes the evolution of bacterial resistance.** *Annu Rev Virol* 2023, **10**:503–524.
 34. Perry EB, Barrick JE, Bohannon BJ: **The molecular and genetic basis of repeatable coevolution between *Escherichia coli* and bacteriophage T3 in a laboratory microcosm.** *PLoS One* 2015, **10**:e0130639.
 35. Meyer JR, et al.: **Repeatability and contingency in the evolution of a key innovation in phage lambda.** *Science* 2012, **335**:428–432.
 36. Thung Tze Y, et al.: **Genetic variation in individuals from a population of the minimalist bacteriophage Merri-merri-uth nyilam marra-natj driving evolution of the virus.** *mBio* 2024, **15**:e02564–24.
 37. Appelmans R: **Le dosage du bactériophage.** *Compt Rend Soc Biol* 1921, **85**:1098.
 38. Maurin A, et al.: **Convergent evolution during host range expansion and virulence increase in a *Salmonella* bacteriophage.** *bioRxiv* 2025, <https://doi.org/10.1101/2024.03.13.584857> p. 2024.03.13.584857.
- This publication evolved five independent populations of a *Tequintavirus* phage against *Salmonella enterica* via the Appelmans protocol. The resulting populations exhibited a broadened host range and increased virulence, and parallel mutations between the populations were detected, indicating repeatable, predictable evolution.
39. Merabishvili M, Pirnay JP, De Vos D: **Guidelines to compose an ideal bacteriophage cocktail.** *Methods Mol Biol* 2024, **2734**:49–66.
 40. Jakob N, et al.: **Appelmans Protocol for in vitro *Klebsiella pneumoniae* phage host range expansion leads to induction of the novel temperate linear plasmid prophage vB_KpnS-KpLi5.** *Virus Genes* 2025, **61**:132–135.

41. Kunisch F, et al.: **Targeting *Pseudomonas aeruginosa* biofilm with an evolutionary trained bacteriophage cocktail exploiting phage resistance trade-offs.** *Nat Commun* 2024, **15**:8572.

In this publication, phages were applied in a modified Appelmans protocol to improve biofilm clearing. A glass bead coated with pre-grown biofilm from the ancestral bacteria was included in each daily setup. Phages demonstrated an improved host range, antimicrobial and anti-biofilm activity, and a two-phage cocktail showed even higher suppression.

42. Laanto E, et al.: **Adapting a phage to combat phage resistance.** *Antibiotics* 2020, **9**:291.
43. Bull JJ, Wichman HA, Krone SM: **Modeling the directed evolution of broad host range phages.** *Antibiotics* 2022, **11**:1709.
44. Pirnay J-P, et al.: **Personalized bacteriophage therapy outcomes for 100 consecutive cases: a multicentre, multinational, retrospective observational study.** *Nat Microbiol* 2024, **9**:1434-1453.

This retrospective examines the first 100 consecutive cases of personalised bacteriophage therapy, looking into the infections treated, phage therapy approaches, and the general efficacy. Notably, some clinical cases utilised phage training to pre-adapt phages against the causative bacterial pathogens.

45. Bonilla N, et al.: **Phage on tap-a quick and efficient protocol for the preparation of bacteriophage laboratory stocks.** *PeerJ* 2016, **4**:e2261.
 46. de Jonge PA, et al.: **Molecular and evolutionary determinants of bacteriophage host range.** *Trends Microbiol* 2019, **27**:51-63.
 47. Peters TL, et al.: **Directed evolution of bacteriophages: thwarted by prolific prophage.** *Appl Environ Microbiol* 2024, **90**:e00884-24.
- This publication used the Appelmans protocol to expand the host range of temperate phages against *Pseudomonas aeruginosa*. Of the 10 output phages studied, those with expanded host range were derived from a prophage likely induced from one of the hosts during the Appelmans protocol.